LUMAFLOW · PROBLEM STATEMENT

LumaFlow problem statement

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	LMF-PRB-001	Version	0.9
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/lumaflow		
Source	docs/01-problem.md @ f88a9c3		

REVISION HISTORY

Version	Date	Author	Change
0.9	2026-10-25	Amish Chadha	Kit 1.8 users-and-needs and stakeholders tables; cost USD 418
0.8	2026-10-02	Amish Chadha	Cost of the constructable design with the 2026-10-02 decisions carried in (USD 416); partner approach status
0.7	2026-10-02	Amish Chadha	Partner as decided on 2026-10-02 (LMF-DEC-001)
0.6	2026-10-01	Amish Chadha	Cost of the constructable design reported against the value-engineering target
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish ($340)
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); dose gap closed at 1.2 L/min, pressure limiter at installation, budget figure open
0.3	2026-09-25	Amish Chadha	Record Amish's decisions (LMF-DDR-001) on the first user, dose target and budget; partner stays open
0.2	2026-09-25	Amish Chadha	

			Populate to TRL 2 (problem, users, context, constraints, prior work with sources)
0.1	2026-09-24	Amish Chadha	Initial scaffold

LumaFlow problem statement

Point-of-use disinfection often depends on mercury UV lamps that break, need a warm-up, run around the clock and are replaced every year, so households with a private well or an unreliable supply either pay for that upkeep or go without. There is no open, inspectable design for a small UV-C LED reactor that switches on only when water flows and tells the user whether the dose was delivered.

The problem

Ultraviolet light at about 250 to 280 nm inactivates bacteria, viruses and protozoa by damaging their DNA and RNA, without adding chemicals or changing taste. Household UV systems have used low-pressure mercury lamps for decades. Those lamps have four drawbacks at the scale of one kitchen tap:

1. **Mercury and glass.** The lamp contains mercury inside a quartz sleeve. A cracked sleeve or lamp can release mercury into the water line and the home. The Minamata Convention on Mercury is driving mercury-containing lamps out of general use (Minamata Convention), and makers of UV-C LED systems market mercury-free operation as the main reason to switch (AquiSense).
2. **Warm-up and continuous running.** A mercury lamp takes time to reach full output, so household units usually run all day and night even though water flows for only a few minutes a day. That wastes energy and heats standing water.
3. **Annual replacement.** Lamp output falls with use, so suppliers advise replacing the lamp about once a year whether or not it has failed (Fresh Water Systems).
4. **Silent failure.** Many low-cost units have no UV sensor. The user cannot tell whether the lamp has aged or the sleeve has fouled. NSF/ANSI 55 Class A systems must include a UV sensor and an alarm, but lower-cost Class B and uncertified units need not (Fresh Water Systems, NSF Class A and B; ANSI blog on NSF/ANSI 55-2024).

UV-C LEDs remove the mercury, reach full output in microseconds and can switch with the flow. Their weakness is efficiency: current UV-C LEDs turn only about 2 to 5 % of electrical input into UV-C, and many lose 30 to 50 % of their output within 5,000 to 10,000 h (Tech-LED selection guide; status of 265 nm LEDs, 2023). A reactor that runs only while water flows turns that weakness around: a few minutes of on-time a day makes the rated life last for many years.

Prior work

- **Commercial UV-C LED units.** AquiSense sells compact flow-through UV-C LED reactors for point-of-use integration (PearlAqua Micro) and a whole-house point-of-entry unit (PearlAqua Deca launch). These are closed designs sold to integrators and households at commercial prices.
- **Research reactors.** Flow-through UV-C LED reactors have been studied for their optics, flow pattern and LED placement (efficiency improvement of a flow-through reactor; cylindrical reactor design optimization; dual-wavelength point-of-use device). Reviews conclude that LED reactors are practical at small flows and that reactor design, not only LED power, sets the delivered dose (critical review, 2024). UV-C LED dose response depends strongly on water quality (UVC-LED efficacy and water quality).
- **Standards and schemes.** NSF/ANSI 55 Class A calls for 40 mJ/cm² at the rated flow and at 70 % UV transmittance or the alarm set point, whichever is lower; Class B calls for 16 mJ/cm² (sources above). The WHO International Scheme to Evaluate Household Water Treatment Technologies rates devices by their log reduction

of bacteria, viruses and protozoa (WHO HWT scheme). The US EPA UV guidance gives dose tables per pathogen; adenovirus in particular needs far higher doses than bacteria (EPA UV toolkit).

The gap LumaFlow addresses is an **open, documented, garage-buildable** tap-scale reactor with flow switching and a dose monitor that fails safe, which researchers, makers and small water projects can inspect, reproduce and test.

Users and context

User	Need	Context
Household on a private well or rainwater tank	Safe drinking water at one tap without chemicals, mercury or a yearly lamp	Under the kitchen sink, feeding a dedicated drinking-water faucet
Household with an intermittent or suspect municipal supply	A barrier against microbial contamination after pressure drops or boil-water events	Same, downstream of a sediment or carbon filter
Small clinic, school or community water point	A low-maintenance disinfection stage for a drinking-water tap	Indoor, mains power available
Researcher or maker	An open reference design to test reactor geometries, LEDs and dose sensing	Lab bench, makerspace

Users and needs

Table 1. Users and needs (kit 1.8).

User type	Goals and needs	Typical use scenario	Pain points and challenges
Household on a private well or rainwater tank (first user, LMF-DDR-001 D9)	Microbiologically safer water at one drinking tap, no chemicals, no mercury, no yearly lamp; know when it is not working	10 to 20 L a day in about 20 short draws from a dedicated faucet; unit under the sink after a sediment filter	Mercury lamps run all day, need warm-up and yearly replacement; low-cost units fail silently; the household is its own utility
Household on an intermittent or suspect municipal supply	A barrier after pressure drops and boil-water events	Same as above, downstream of a carbon or sediment filter	Contamination events are unpredictable; boiling is a chore
Small clinic, school or community water point	A low-maintenance disinfection stage for one drinking tap, with a visible fault signal	Indoor tap on mains power, used by many people a day	No technician on site; parts must be replaceable locally
Researcher or maker	An open, documented reference reactor to test optics, LEDs and dose sensing	Lab bench or makerspace	Commercial units are closed and cannot be inspected or modified

Stakeholders

Table 2. Stakeholders.

Stakeholder group	Role or interest	Engagement needed
University extension programme for private well owners (Texas A&M AgriLife Extension Texas Well Owner Network as first candidate, LMF-DEC-001)	Samples household wells; can measure UV transmittance and microbiology on real water	First approach by Amish (not yet made); co-design of requirements and TRL 4 testing
Certification bodies (NSF, a certified laboratory)	Own NSF/ANSI 55 and the material standards a product would need	Informative only at TRL 3; certification is out of scope in this phase
Local plumbers and installers	Fit the pre-filter, pressure limiter and push-fit lines	Review of the installation requirements (PR-017)
Makerspaces and university labs	Build and test copies of the open design	Build reports through the issue templates once the repo is public
Component suppliers (UV-C LEDs, PTFE reflector, quartz)	Datasheets for reflectance, output bin and lifetime	Data requests at part selection (decisions register, To confirm)

Typical conditions assumed for the concept: water supply pressure of 2 to 6 bar (29 to 87 psi), water temperature 5 to 30 °C, cabinet air up to 35 °C, and a drinking and cooking demand of about 10 to 20 L per household per day drawn in short bursts.

Constraints

- Garage-buildable prototype, $340 USD or less in parts (budget raised from $200 to $225 by Amish, LMF-DDR-001 D2, then to $340 by the budget top-up Amish approved on 2026-09-26), from off-the-shelf LEDs, sensors and plumbing plus simple machined end caps. The $340 figure is a value-engineering target (Amish, 2026-10-01); the constructable design of LMF-DDR-003, with the decisions of 2026-10-02 carried in, is estimated at $433.00, $93 over it.
- A drinking-water flow of 1.2 L/min (0.32 gpm) is enough for one tap; it fills a 1 L bottle in about 50 s.
- A sediment pre-filter and a pressure-limiting valve at 4 bar (58 psi) or less are part of the installation.
- Fits under a kitchen sink and connects to a standard cold line with push-fit fittings.
- Wetted parts must be food-contact grade; no printed plastic on the water path.
- Only safety extra-low voltage (24 V DC) at the unit; mains stays inside a certified external adapter.
- Research and educational prototype only. LumaFlow is not a certified water treatment device, and the documents must not claim that its water is safe to drink.

Out of scope

- Removal of chemicals, metals, nitrate, taste or odor. UV does not remove them, and it leaves no residual disinfectant in the pipes after the reactor.
- Whole-house (point-of-entry) flow rates of 20 L/min or more.
- Turbid or iron-rich water without pre-treatment; UV needs clear water to work.
- Certification testing to NSF/ANSI 55 or the WHO scheme (a later step, not part of this portfolio phase).

Decisions and open questions

- **First user (decided).** A well-water household with a sediment pre-filter. Decided by Amish, 2026-09-25: go with recommendation (LMF-DDR-001 D9). This sets the design water quality at 90 %/cm UVT or better.
- **Dose target (decided).** The Class A dose level of 40 mJ/cm² is the target for clear water (90 %/cm or better), with an alarm and valve closure below it, and the 70 %/cm test condition kept visibly not met. Decided by Amish, 2026-09-25: go with recommendation (LMF-DDR-001 D1). The first TRL 3 calculation found only 14.7 to 19.0 mJ/cm² in clear water; with the route Amish accepted on 2026-09-25 (LMF-DDR-002: 50 mm bore, high-reflectance liner, 1.2 L/min design flow) LMF-CAL-001 v0.2 gives 51.9 to 76.3 mJ/cm².
- **Partner (decided).** A university extension programme for private well owners, which already samples household wells and can measure UV transmittance on real samples; the Texas A&M AgriLife Extension Texas Well Owner Network is the first candidate to approach. Decided by Amish, 2026-10-02 (LMF-DEC-001). No approach has been made yet and nothing is agreed; a draft first message for Amish to send is in the review note (2026-10-02).